New bounds for small two- and four-colour Ramsey numbers

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Abstract

We give upper bounds for the Ramsey numbers $\mathrm{R}(3,k)$ with $24 \le k \le 50$, for which Radziszowski's survey lists no upper bounds. They come from an exact bootstrap of lower bounds on the number of edges of triangle-free graphs with bounded independence number. It starts from the published values and lower bounds of Goedgebeur and Radziszowski. For instance, $\mathrm{R}(3,24) \le 201$, $\mathrm{R}(3,30) \le 297$ and $\mathrm{R}(3,50) \le 729$. With in addition the table of Backelin's unrefereed report, the bounds are 1 to 3 lower for $24 \le k \le 36$ except $k = 35$.

Deleting vertices from the quartic residue graphs of Lindsay and Cain gives $\mathrm{R}(4,21) \ge 260$ and $\mathrm{R}(4,24) \ge 370$. Finally, we use the four-colour form of the neighbourhood recurrence, with $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$ from a companion paper and the survey's $\mathrm{R}(3,3,3,4) \le 149$. This gives $\mathrm{R}(3,3,4,4) \le 442$, $\mathrm{R}(3,4,4,4) \le 1521$ and $\mathrm{R}(4,4,4,4) \le 6082$. All certificates and witnesses are checked exactly by programs in the ancillary files.

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1 Introduction

For integers $s,t \ge 1$, the Ramsey number $\mathrm{R}(s,t)$ is the least n such that every red-blue colouring of the edges of K_n contains a red K_s or a blue K_t; equivalently, every graph on n vertices contains a K_s or an independent set of size t. Multicolour numbers $\mathrm{R}(t_1,\dots,t_r)$ are defined in the same way. The current state of knowledge is collected in Radziszowski's dynamic survey [12], whose revision 18 we use for all comparisons, together with the later preprint [16].

This paper is a companion to a study of three-colour numbers [4], in which local Ramsey bounds inside colour neighbourhoods are combined with double counting and with flag-algebra certificates. The present paper gives new bounds for two-colour numbers. The upper bounds are obtained by adapting the ladder procedure of [4, Section 5], and the lower bounds by deleting vertices from quartic residue graphs. It also derives consequences for four-colour numbers. Our results are listed in Table 1.

The upper bounds for $\mathrm{R}(3,k)$ (Section 2) and the lower bounds for $\mathrm{R}(4,k)$ (Section 3) are computer-assisted. Floating-point solvers and local search were used only to find certificates

number	previous	new	where
$\mathrm{R}(3,k)$, $24 \le k \le 50$	no upper bound [12]	Table 2	Theorem 2.1
$\mathrm{R}(4,21)$	≥ 242 [15]; ≥ 257 ([16] and [12, 2.3(e)])	≥ 260	Theorem 3.1
$\mathrm{R}(4,24)$	≥ 343 (implicit in [11])	≥ 370	Theorem 3.1
$\mathrm{R}(3,3,4,4)$	≤ 450 [12]	≤ 442	Theorem 4.1
$\mathrm{R}(3,4,4,4)$	≤ 1576 [12] (implicit in [2])	≤ 1521	Theorem 4.1
$\mathrm{R}(4,4,4,4)$	≤ 6306 [12]; ≤ 6301 (implicit in [2])	≤ 6082	Theorem 4.1

Table 1: Our results. For $\mathrm{R}(3,k)$ with $k \ge 24$, iterating $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1)+k$, which is strict when both terms are even, or Shearer's bound [14], gives weaker explicit bounds (Table 2). Where the two rows of Table 2 differ, the smaller bound also uses the unrefereed report [1].

and witnesses, and in the negative experiments that we report. Every certificate and witness is checked in exact arithmetic by a program that does not share code with the search (Section 5).

The four-colour bounds (Section 4) follow by hand from $\mathrm{R}(3,3,3,4) \le 149$ [12] and from $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$, which are proved in [4, Theorems 8.3 and 8.2] by certificates checked in exact arithmetic by computer.

2 Upper bounds for $\mathrm{R}(3,k)$

A $(3,k;n)$-graph is a triangle-free graph on n vertices with no independent set of size k. A $(3,k;n,e)$-graph is one with e edges, and $e(3,k,n)$ is the least such e. As in [4, Section 3], $m_k(n)$ is a *valid lower bound* if every $(3,k;n)$-graph has at least $m_k(n)$ edges.

For a vertex v of a $(3,k;n,e)$-graph G, $N(v)$ is independent, so $d_v \le k-1$. Moreover, $G_v = G - N[v]$, where $N[v] = N(v) \cup \{v\}$, is a $(3,k-1;n-1-d_v)$-graph, as an independent set of G_v extends by v. As no edge lies inside $N(v)$, G_v has $e - Z_v$ edges, where

$$Z_v = \sum_{u \in N(v)} d_u.$$

As $\sum_v Z_v = \sum_v d_v^2$, valid lower bounds m_{k-1} give

$$\sum_v \left(d_v^2 + m_{k-1}(n-1-d_v) \right) \le ne.$$

This is the inequality of Graver and Yackel [8], on which Goedgebeur and Radziszowski [6] base their bounds for $k \ge 11$. It is the two-colour analogue of the density inequality [4, Lemma 3.3]. As $ne = \sum_v nd_v/2$, it says

$$\sum_v \varphi(d_v) \le 0 \qquad \text{where} \qquad \varphi(d) = d^2 + m_{k-1}(n-1-d) - nd/2.$$

The bootstrap. We bootstrap as in the ladder of [4, Section 5], with this inequality in place of the density inequality. Level $k \ge 11$ uses the bound B_{k-1} on $\mathrm{R}(3,k-1)$ and the valid lower bounds m_{k-1} from level $k-1$. Level 11 starts from $B_{10} = 42$ [6, Theorem 6] and the values and lower bounds for $e(3,10,n)$ in [6].

Level k treats the orders $n = 0, 1, 2, \ldots$ in turn, records a valid lower bound $m_k(n)$, and stops at the first order it excludes, B_k. So $\mathrm{R}(3,k) \le B_k$. At order n the procedure is as follows.

(i) The admissible degrees are the $d \le \min(k-1, n-1)$ with $n-1-d \le B_{k-1}-1$.

(ii) An order $n \ge 1$ is excluded if $\varphi(d) > 0$ at every admissible d (in particular if there is none), as $\sum_v \varphi(d_v)$ would then be a sum of n positive terms.

(iii) Otherwise let e_0 be the largest of the following bounds. The first is the bound of Radziszowski and Kreher [6, Theorem 2]: $e(3,k,n) \ge 6n - 13(k-1)$ for all n, with exact piecewise-linear values for $n \le 13(k-1)/4 - 1$. If the lower convex hull of the points $(d, \varphi(d))$, d admissible, falls to 0 along an edge, the second is $\lceil n\alpha/(2\beta) \rceil$, where $\alpha - \beta d$ is the line through that edge. This line has $\beta > 0$ and lies below φ at every admissible d, so

$$0 \ge \sum_v \varphi(d_v) \ge n\alpha - 2\beta e.$$

(iv) The extra step below, at the orders where the search applied it (each is recorded in the certificates), may exclude e_0 and some larger values of e. The recorded $m_k(n)$ is the least $e \ge e_0$ that it does not exclude. If it excludes every e from e_0 to $\lfloor n(k-1)/2 \rfloor$, the most edges possible when every degree is at most $k-1$, then n is excluded.

In the chain with [1], the lower bounds of [1] on $e(3,k,n)$ for $11 \le k \le 13$ and $35 \le n \le 43$ are a further candidate for e_0 in (iii); at $k = 10$ they agree with those of [6]. Deleting a vertex as in [4, Lemma 3.5(a)], which gives $m_k(n-1) + \max\{0, n - B_{k-1}\}$, never does better.

An extra step. We add one step, for $e = e_0, e_0+1, \ldots$ in turn. We enumerate every degree sequence with sum $2e$ and admissible entries, each at most $e - m_k(n-1)$, that satisfies the Graver–Yackel inequality. The cap on the entries holds as deleting v leaves a $(3,k;n-1)$-graph with $e - d_v$ edges.

We then try to exclude each sequence by a small linear programme. The *class* of an entry a is the set of the c_a vertices of degree a. The variables are the numbers $x_{ab} = x_{ba}$ of edges between the classes of $a \ne b$ and the numbers x_{aa} of edges inside a class. Counting edge ends, write $x_{a\to b} = x_{ab}$ for $a \ne b$ and $x_{a\to a} = 2x_{aa}$, so that $\sum_{d_v=a} Z_v = \sum_b b\, x_{a\to b}$. With $z_a = e - m_{k-1}(n-1-a)$, every $(3,k;n,e)$-graph with that degree sequence satisfies, for all entries a and b:

- $\sum_b x_{a\to b} = a\, c_a$;
- $\sum_b b\, x_{a\to b} \le c_a z_a$, as G_v has $e - Z_v \ge m_{k-1}(n-1-a)$ edges;
- $\sum_b b\, x_{a\to b} \ge c_a\big(e - \lfloor (n-1-a)(k-2)/2 \rfloor\big)$, as G_v is a $(3,k-1)$-graph and so has maximum degree at most $k-2$;
- $x_{ab} \le c_a c_b$ for $a \ne b$;
- $\sum x_{ab} \le \lfloor N_S^2/4 \rfloor$, summed over $a \le b$ in S, for every set S of classes with N_S vertices in all (Mantel's theorem);
- $\sum_{b' \le b} x_{a\to b'} \ge t\, c_a$, with $t = t(a,b)$ as follows.

For the last constraint, let q be the least entry above b, and let v be a vertex of degree a. For $0 \le j \le a$, let $L(j)$ be the sum of the j smallest entries at most b, other than the entry of v,

plus $(a-j)q$. Put $L(j)=\infty$ if there are fewer than j such entries, or if no entry exceeds b and $j<a$. If v has exactly j neighbours of degree at most b, then $L(j)\le Z_v\le z_a$ (so no such v exists when $L(j)=\infty$). So v has at least t such neighbours, where t is the least j with $L(j)\le z_a$, and summing over the class of a gives the last constraint. If there is no such j, no vertex of degree a exists and the sequence is excluded by this *neighbour-count test*.

If every sequence is excluded, so is e, and the step moves on to $e+1$. Each exclusion is certified either by the neighbour-count test or by rational Farkas multipliers, which combine these constraints, with $x\ge 0$, into an inequality $0\ge c$ with $c>0$.

Theorem 2.1 (computer-assisted)**.** *From the published values and lower bounds of [6, Theorem 2 and Tables 3–5] and* $\mathrm{R}(3,10)\le 42$ *[6, Theorem 6],* $\mathrm{R}(3,k)$ *is at most the value in the row "[6] only" of Table 2 for* $24\le k\le 50$*. With in addition the lower bounds on* $e(3,k,n)$ *in the table of [1, Section 5], it is at most the value in the row "with [1]".*

k	24	25	26	27	28	29	30	31	32
[6] only	201	216	231	247	264	280	297	315	333
with [1]	200	215	230	245	261	278	295	312	331
k	33	34	35	36	37	38	39	40	41
[6] only	351	370	389	409	429	450	471	492	513
with [1]	349	369	389	408	429	450	471	492	513
k	42	43	44	45	46	47	48	49	50
both	536	559	582	606	630	654	679	704	729

Table 2: Upper bounds for $\mathrm{R}(3,k)$ from Theorem 2.1. The two rows coincide for $k=35$ and $k\ge 37$, and for $k\ge 42$ they are shown as one, "both". The survey [12] lists no upper bounds for these k. For comparison, $\mathrm{R}(3,k)\le \mathrm{R}(3,k-1)+k$, strict when both terms are even, iterated from the survey's $\mathrm{R}(3,23)\le 185$, gives 209, 234, 259, 371 and 1171 for $k=24,25,26,30,50$. Shearer's bound [14] $\mathrm{R}(3,k)\le \lfloor (k-1)/f(k-1)\rfloor+1$, with $f(d)=(d\ln d-d+1)/(d-1)^2$, gives 223 for $k=24$ and 792 for $k=50$.

The two chains. Above level 10, each chain uses only the bounds B_{k-1} that it proves itself, never the unpublished values of [12]. Neither chain needs the bound $\mathrm{R}(3,10)\le 41$ listed in [12]. The report [1] is a preprint that gives little of its proofs, so the second row rests on inputs without published proofs. The first row is 1 to 3 larger for $24\le k\le 34$ and $k=36$, and equals the second for $k=35$ and $k\ge 37$.

Without the extra step, the chain from [6] alone gives the same bounds for $11\le k\le 50$, except $\mathrm{R}(3,20)\le 146$ and $\mathrm{R}(3,35)\le 390$. This run is not among the certificates in the ancillary files. The chain from [6] alone also raises seven lower bounds of [6, Tables 9–11] by one, to

$$e(3,13,58)\ge 261,\qquad e(3,14,69)\ge 366,\qquad e(3,14,70)\ge 382,$$
$$e(3,15,n)\ge 516,\ 534,\ 553 \text{ and } 573 \quad \text{for } n=82,\dots,85.$$

Backelin's unpublished tables may already contain these.

Checking the certificates. The certificates are checked by a program that does not import the generating code. It enumerates the degree sequences itself, rebuilds each constraint from its definition, and verifies the multipliers in exact arithmetic.

- The chain with [1] has one row for each level $11 \le k \le 50$ and each order $0 \le n < B_k$, 13,376 rows in all. It excludes 765 degree sequences at 53 triples (k, n, e), 686 by the neighbour-count test and 79 by Farkas multipliers.
- The chain from [6] alone has 13,403 rows and excludes 766 sequences at 50 triples, 684 and 82 respectively.

Comparison with the survey. For $k \le 23$ the procedure improves no value of [12]. From [6] alone it gives $\mathrm{R}(3, 17) \le 109$ and $\mathrm{R}(3, 20) \le 145$, which [12] credits to an unpublished manuscript of Backelin and to Lesser's thesis [10], respectively. With the table of [1], it lowers the bounds $\mathrm{R}(3, 16) \le 98$ and $\mathrm{R}(3, 19) \le 133$ that it gives from [6] alone to 97 and 132, the values stated in [1].

With the table of [1] it also gives $\mathrm{R}(3, 23) \le 185$, against 187 from [6] alone. The survey [12] credits this bound to a personal communication of Backelin; it now has a derivation that can be checked, given the inputs of [1]. At $k = 18$, 21 and 22 this chain gives 121, 158 and 172, one more than the values 120, 157 and 171 that [12] lists from unpublished sources (from [6] alone, 121, 159 and 173).

For $k = 18$, both chains give $m_{18}(120) = 1020 = 120 \cdot 17/2$ from the hull. As every degree is at most 17, a $(3, 18; 120)$-graph would have exactly 1020 edges and be 17-regular. Then every G_v would be a $(3, 17; 102, 731)$-graph, as $120 - 1 - 17 = 102$ and $1020 - 17^2 = 731$. So $e(3, 17, 102) \ge 732$, one more than our $m_{17}(102) = 731$, would exclude $n = 120$ and give $\mathrm{R}(3, 18) \le 120$.

3 Lower bounds for $\mathrm{R}(4, k)$

For a prime $p \equiv 1 \pmod 8$ let Q_p be the circulant graph on $\mathbb{Z}_p$ in which $x \sim y$ if and only if $x - y$ is a nonzero fourth power. The relation is symmetric, as -1 is a fourth power. Lindsay and Cain [11] showed that Q_{313} and Q_{457} are K_4-free with no independent sets of size 22 and 25, respectively. Their independence numbers are

$$\alpha(Q_{313}) = 21 \qquad \text{and} \qquad \alpha(Q_{457}) = 24.$$

The upper bounds are those of [11]. The ancillary checks confirm $\alpha(Q_{313}) = 21$ and show that Q_{457} has an independent 24-set.

If a vertex set S meets every independent set of size $\alpha(Q_p)$, then $Q_p - S$ is K_4-free and has independence number less than $\alpha(Q_p)$.

Theorem 3.1 (witnesses checked by computer)**.** $\mathrm{R}(4, 21) \ge 260$ *and* $\mathrm{R}(4, 24) \ge 370$.

Proof. Step 1: symmetries and cosets. The maps $x \mapsto ax + b$ with a a nonzero fourth power are automorphisms of Q_p. Let O_1, O_2, O_3 be the cosets of the group of nonzero fourth powers in $\mathbb{Z}_p^*$ other than the group itself, in a fixed order, and let r_j be the least element of O_j.

Step 2: representatives. Let $k \geq 2$, let J be an independent k-set and let $u \in J$. The translate $J - u = \{x - u : x \in J\}$ is an independent k-set that contains 0, so its other elements are not fourth powers and lie in $O_1 \cup O_2 \cup O_3$. Let j be least with $J - u$ meeting O_j, say at w. Then r_j/w is a fourth power, and multiplying by it maps $J - u$ to an independent k-set that contains 0 and r_j and still avoids $O_1, \ldots, O_{j-1}$, as multiplication by a fourth power fixes each coset. Call an independent k-set a *representative* if, for some j, it contains 0 and r_j and avoids $O_1, \ldots, O_{j-1}$. So every independent k-set is the image of a representative under one of the maps above.

Step 3: the computation. An exhaustive search lists all representatives for $(p, k) = (313, 21)$ and $(457, 24)$. A weighted local search found sets S of 54 and 88 vertices, listed in the ancillary files, and an exact program checks that each meets every image of every representative. So $Q_{313} - S$ is a K_4-free graph on 259 vertices with no independent 21-set, and $Q_{457} - S$ one on 369 vertices with no independent 24-set. □

Verification. Both graphs are rebuilt from the lists of deleted vertices by programs that share no code with the search. The 259-vertex graph is also checked by a direct exhaustive search for K_4 and for independent 21-sets, and the 369-vertex graph by one for K_4 and for independent 24-sets. Neither search uses symmetry, so each is independent of the orbit argument above. □

Comparison with earlier bounds. The previous bounds are as follows.

- For $\mathrm{R}(4, 21)$, the survey has 242 [15], and $\mathrm{R}(4, 20) \geq 252$ from a recent preprint [16] gives 257 by the inequality $\mathrm{R}(4, l) \geq \mathrm{R}(4, l - 1) + 5$ [12, 2.3(e)].
- For $\mathrm{R}(4, 24)$, the non-neighbourhood of a vertex of Q_{457} (Q_{457} is 114-regular) has 342 vertices, is K_4-free and has no independent 24-set, as such a set would extend by the vertex. This gives $\mathrm{R}(4, 24) \geq 343$. The survey lists no value for $\mathrm{R}(4, 24)$. Its rules give 324, by applying this inequality twice to $\mathrm{R}(4, 22) \geq 314$ [11].

The method does not reach $\mathrm{R}(4, 23) \geq 320$, which would improve the bound 319 that this inequality gives from $\mathrm{R}(4, 22) \geq 314$. That needs a set of at most $457 - 319 = 138$ vertices meeting every independent 23-set of Q_{457}, and the smallest we found has 147.

4 Four colours

For $\mathbf{t} = (t_1, \ldots, t_4)$ and the unit vectors $\mathbf{e}_i$, the four-colour form of the neighbourhood recurrence [12, 6.1(a)] is

$$\mathrm{R}(\mathbf{t}) \leq \sum_{i=1}^{4} \mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 2.$$

Indeed, consider a colouring of K_n with no colour-j K_{t_j} for any j. The colour-i neighbourhood of a vertex v contains no colour-i $K_{t_i - 1}$, which would extend by v, and no colour-j K_{t_j} for $j \neq i$. So it has at most $\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1$ vertices. Summing over i gives

$$n - 1 \leq \sum_i (\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1).$$

The recurrence improves to $\mathrm{R}(\mathbf{t}) \le \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$ when $\sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even and some $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even. Indeed, take a colouring as above of K_n with $n = \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$, which is odd. Then the display holds with equality, so every vertex has colour-i degree exactly $\mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 1$ for each i. For an i with $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ even this degree is odd, and a graph on an odd number of vertices cannot have all its degrees odd.

Both statements hold with the same proofs when each $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is replaced by a proven upper bound for it. An entry equal to 2 may be dropped, leaving a number with fewer colours, as a colouring with no colour-i K_2 has no colour-i edges.

Theorem 4.1. $\mathrm{R}(3,3,4,4) \le 442$, $\mathrm{R}(3,4,4,4) \le 1521$ *and* $\mathrm{R}(4,4,4,4) \le 6082$.

Proof. With $\mathrm{R}(3,4,4) \le 73$ [4, Theorem 8.3] and $\mathrm{R}(3,3,3,4) \le 149$ [12], the recurrence gives

$$\mathrm{R}(3,3,4,4) \le 2 \cdot 73 + 2 \cdot 149 - 2 = 442.$$

Parity does not apply, as every term is odd.

With $\mathrm{R}(4,4,4) \le 198$ [4, Theorem 8.2] it then gives

$$\mathrm{R}(3,4,4,4) \le 198 + 3 \cdot 442 - 2 = 1522.$$

This is even, as are all four terms, so $\mathrm{R}(3,4,4,4) \le 1521$.

Finally,

$$\mathrm{R}(4,4,4,4) \le 4 \cdot 1521 - 2 = 6082,$$

and parity does not apply, as 1521 is odd. □

The inputs and earlier bounds. The two inputs from [4] are computer-assisted. The survey's $\mathrm{R}(3,3,3,4) \le 149$ is itself the recurrence with parity, $3 \cdot 30 + 62 - 2 = 150$, from $\mathrm{R}(3,3,4) = 30$ [3] and $\mathrm{R}(3,3,3,3) \le 62$ [5]. From the survey's $\mathrm{R}(3,4,4,4) \le 1576$, the recurrence with parity gives $\mathrm{R}(4,4,4,4) \le 4 \cdot 1576 - 3 = 6301$, as $4 \cdot 1576$ and all four terms are even. The survey lists 6306 and notes that 6301 is implicit in [2].

With the survey's $\mathrm{R}(3,3,4,4) \le 450$ in place of 442, that is, without $\mathrm{R}(3,4,4) \le 73$, the bound $\mathrm{R}(4,4,4) \le 198$ alone gives $\mathrm{R}(3,4,4,4) \le 1545$ and $\mathrm{R}(4,4,4,4) \le 6178$. We have not tried the analogue of the parity clause modulo 3 of [2], which gives the survey's $\mathrm{R}(3,4,4,4) \le 1576$.

Four triangle-free colours. We also tried to improve $\mathrm{R}(3,3,3,3) \le 62$ [5], that is, to exclude colourings of K_{61} with four triangle-free colours, by certificates analogous to those for $\mathrm{R}(4,4,4)$ in [4, Sections 7 and 8]. Such a certificate exists only if a flag relaxation, here with four- or five-vertex patterns and with four-colour analogues of the local caps of [4, Lemmas 2.1 and 2.3] among its constraints, is infeasible. As in Razborov's flag algebras [13], the unknowns of the relaxation are the densities of the colourings of K_4 or K_5, the patterns, in a putative colouring of K_{61}. They are subject to linear constraints from the caps, such as $|N_i(v) \cap N_j(u)| \le \mathrm{R}(3,3) - 1 = 5$ for $u \in N_i(v)$ and $j \ne i$, and to positive-semidefiniteness constraints. A certificate is a dual solution that proves the relaxation infeasible.

At $n = 61$ every colour degree is at most $\mathrm{R}(3,3,3) - 1 = 16$ and the four sum to 60. So the *profile* of a vertex, its sorted quadruple of colour degrees, is one of

$$(12,16,16,16), \quad (13,15,16,16), \quad (14,14,16,16), \quad (14,15,15,16), \quad (15,15,15,15).$$

Both relaxations are numerically feasible. Floating-point solvers found an assignment of pattern densities meeting all their constraints, even when every vertex is required to have profile $(12, 16, 16, 16)$. The four-vertex relaxation was solved as a linear programme by HiGHS [9], with its semidefinite constraint imposed by eigenvector cuts until no eigenvalue was below -10^{-8}. The five-vertex relaxation was solved as a semidefinite programme by Clarabel [7] at tolerance 10^{-8}. So we obtained no such certificate from these relaxations, and no bound follows.

5 Verification, literature and limitations

Ancillary files. This paper and [4] have the same ancillary files, which are also in the directory `ancillary/` of the repository

`https://github.com/Lachy-Dauth/ramsey-neighbourhood-counting`.

They contain the programs and data, among them:

- `two_colour_r3k/` for Theorem 2.1, with the inputs from [6] and [1] and the certificates of both chains;
- `two_colour_r4k/` for Theorem 3.1, with the lists of deleted vertices, `build_power_sub.py`, which rebuilds the two graphs from them, the exact checkers `rcheck.c` (direct search) and `qpsym.c` (orbit argument), and `verify_all.sh`, which checks the SHA-256 hashes of the two graphs and runs the checkers;
- `flag5_r444/` and `flag5_r34k/`, which check the inputs $\mathrm{R}(4,4,4) \le 198$ and $\mathrm{R}(3,4,4) \le 73$ of Theorem 4.1 from [4].

The command `sh run_all.sh` runs the quick checks of both papers in a few minutes. The command `sh run_all.sh --full` adds the long ones, among them the checks of these two inputs and the direct search of the 369-vertex graph for independent 24-sets.

The checkers are written in Python and C with their standard libraries. The C programs use POSIX threads. Negative controls (a weakened input, a removed multiplier or sequence certificate, a value raised by one, a forged cap, a deleted vertex put back) are all rejected.

Literature. We searched arXiv and the web up to 23 October 2026 for earlier statements of our new bounds, and found none.

- For $\mathrm{R}(3,k)$ with $k \ge 24$ the survey lists no upper bounds, and the upper bounds of [6] and of Lesser's thesis [10] stop at $k = 16$ and $k = 21$. Backelin's unpublished manuscript, to which [12] credits $\mathrm{R}(3,17) \le 109$, may contain bounds of this kind.
- For $\mathrm{R}(4,21)$ the best previous bound we know of, 257, follows from $\mathrm{R}(4,20) \ge 252$ in the preprint [16], which we have not checked. Without it the bound is 242 [15].
- For $\mathrm{R}(4,24)$ the survey lists nothing, and 343 is implicit in [11].
- For the four-colour numbers the previous bounds are those of [12] and [2] in Table 1.

Such a search cannot establish priority, and we would welcome pointers to any earlier source.

Limitations. For $k \le 23$ the bootstrap of Section 2 improves none of the bounds in [12]. At $k = 18$ it would give $\mathrm{R}(3, 18) \le 120$ if its own lower bound $e(3, 17, 102) \ge 731$ were one larger. The second row of Table 2 rests on the lower bounds of [1], whose proofs are largely unpublished.

The deletion method of Section 3 does not reach $\mathrm{R}(4, 23) \ge 320$. The four-colour bounds rest on $\mathrm{R}(3, 4, 4) \le 73$ and $\mathrm{R}(4, 4, 4) \le 198$ [4, Theorems 8.3 and 8.2], which are computer-assisted, and on $\mathrm{R}(3, 3, 3, 4) \le 149$, which [12, 6.1(a)] derives from $\mathrm{R}(3, 3, 4) = 30$ [3] and $\mathrm{R}(3, 3, 3, 3) \le 62$ [5], both computer-assisted.

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References

[1] J. Backelin, Edge number report 1: state of the art estimates for $n \le 43$, preprint, arXiv:1410.1843, 2014.

[2] L. Boza, New upper bounds for the classical Ramsey numbers $R(4, 4, 4)$, $R(3, 4, 5)$ and $R(3, 3, 6)$, arXiv:2603.10851, 2026.

[3] M. Codish, M. Frank, A. Itzhakov and A. Miller, Computing the Ramsey number $R(4, 3, 3)$ using abstraction and symmetry breaking, *Constraints* 21 (2016), 375–393.

[4] L. Dauth, Neighbourhood counting and new bounds for small three-colour Ramsey numbers, companion paper, 2026. The theorem and section numbers cited are those of its first arXiv version.

[5] S. E. Fettes, R. L. Kramer and S. P. Radziszowski, An upper bound of 62 on the classical Ramsey number $R(3, 3, 3, 3)$, *Ars Combin.* 72 (2004), 41–63.

[6] J. Goedgebeur and S. P. Radziszowski, New computational upper bounds for Ramsey numbers $R(3, k)$, *Electron. J. Combin.* 20(1) (2013), #P30.

[7] P. J. Goulart and Y. Chen, Clarabel: an interior-point solver for conic programs with quadratic objectives, preprint, arXiv:2405.12762, 2024.

[8] J. E. Graver and J. Yackel, Some graph theoretic results associated with Ramsey's theorem, *J. Combin. Theory* 4 (1968), 125–175.

[9] Q. Huangfu and J. A. J. Hall, Parallelizing the dual revised simplex method, *Math. Program. Comput.* 10 (2018), 119–142.

[10] A. Lesser, Theoretical and computational aspects of Ramsey theory, Examensarbeten i Matematik 3, Stockholms Universitet, 2001.

[11] M. Lindsay and J. W. Cain, Improved lower bounds on the classical Ramsey numbers $R(4, 22)$ and $R(4, 25)$, preprint, arXiv:1510.06102, 2015.

[12] S. P. Radziszowski, Small Ramsey numbers, *Electron. J. Combin.*, Dynamic Survey DS1, revision 18, 24 April 2026.

[13] A. A. Razborov, Flag algebras, *J. Symbolic Logic* 72 (2007), 1239–1282.

[14] J. B. Shearer, A note on the independence number of triangle-free graphs, *Discrete Math.* 46 (1983), 83–87.

[15] W. Su, H. Luo, Z. Zhang and G. Li, New lower bounds of fifteen classical Ramsey numbers, *Australas. J. Combin.* 19 (1999), 91–99.

[16] C. Yu, Ramsey number $R(4, 20) \geq 252$, preprint, arXiv:2608.18169, 2026.